

Event-Level Spectrogram Object Detection and Curation for Avian Acoustic Analysis

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Abstract

Passive acoustic monitoring can capture large quantities of bird audio, but answering some ecological questions about species with short call types requires localized timing and frequency-band information beyond clip-level presence labels alone. We present a two-stage computer-vision pipeline on spectrograms, demonstrated on Hume’s Leaf Warbler buzz vocalizations and designed for short, spectrally localized acoustic events. Stage A uses object detection to localize buzzes in time-frequency spectrogram windows and returns event confidence, start/end times, and frequency bounds; detections from overlapping windows are then consolidated into event-level records. Stage B classifies whether the predicted buzz event represented by each crop is complete or incomplete to curate truncated or masked events before duration analysis. We evaluate Stage A against full-window classifier baselines, including AST and YOLO-classification, and show that object detection remains competitive under shared window-level scoring while uniquely providing event timing and frequency localization. Native event-level evaluation further shows strong retrieval performance for completely observed buzzes. Finally, we illustrate how curated event records can be exported as analysis-ready summaries for ecological workflows, including duration and frequency-localization data products.

Keywords: Bioacoustics, Passive Acoustic Monitoring, Object Detection, Event Localization, Spectrogram Classification, Hume’s Leaf Warbler

1. Introduction

Passive acoustic monitoring (PAM) is widely used for ecological observation, including biodiversity monitoring, phenology tracking, and habitat-use

assessment, because it can collect long-duration audio across space and time with low disturbance (Sugai et al., 2019; Kahl et al., 2021). In practice, many PAM machine-learning pipelines return clip-level presence/absence or activity scores over fixed windows. These outputs are useful for occupancy and broad activity trends, but many ecological questions require event-level quantities such as call rate, event duration, inter-event spacing, and frequency-band use across sites or seasons. For call types where event timing and frequency localization are ecological signals of interest, clip-level predictions alone do not provide the measurements needed for downstream analysis.

The Hume’s Leaf Warbler (HLW), or *Phylloscopus humei*, a high-elevation Himalayan passerine, is one such case: males produce a distinctive high-frequency buzz vocalization (Fig. 1) during the breeding season, but event-level acoustic measurements for this call type remain limited. Analyses of buzz structure and potential ecological correlates require event-level quantities, including start time, duration, and frequency-band placement. In field recordings, these buzzes are embedded in variable background noise, and incompletely observed buzzes commonly arise under low SNR (e.g., distant birds) and transient non-stationary noise (Fig. 3). Buzzes can also overlap with other acoustic activity, including other HLW buzzes (Fig. 4). For this measurement goal, the analysis pipeline must localize buzzes in time-frequency space, preserve distinct overlapping events where possible, and flag incompletely observed events before summary analysis. Otherwise, truncated or masked buzzes can underestimate event duration or frequency extent and bias downstream summaries.

Our ecological motivation for studying the Hume’s Leaf Warbler buzz vocalization was to test the hypothesis that attributes of this vocalization vary systematically across the studied elevational gradient (~ 3200 – 3800 m). Environmental conditions within this narrow range vary dramatically. Lower elevations within this range exhibit much earlier tree leaf-out and harbor more abundant insect communities than upper elevations. We therefore hypothesized that more resource-rich lower elevations are better environments for breeding Hume’s Leaf Warblers than upper elevations. Male Hume’s Leaf Warblers produce buzzes during the breeding season to secure breeding territories and mates: the exact function of this vocalization remains uncertain. We examine two testable predictions that derive from the hypothesis that lower elevations are preferred over upper elevations. First, we predicted that buzz vocalizations should occur more frequently at lower elevations, as we expect greater interaction rates (between rival territorial males and/or be-

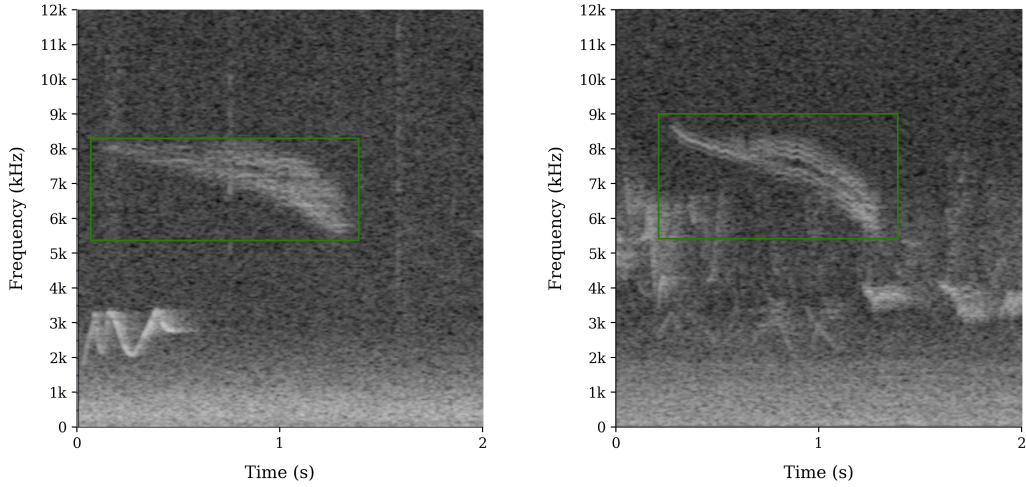


Figure 1: Representative HLW buzz spectrograms from initial data exploration.

tween territorial males and females) in this zone. Second, we predicted that buzz vocalizations should be longer at lower elevations, as we expect males at lower elevations to be in better condition and therefore able to produce more exaggerated displays (e.g. longer buzzes, which presumably are more energetically demanding to produce).

Testing these ecological predictions requires a computer-vision formulation on spectrograms, where each buzz is treated as a localized object with explicit temporal and spectral extent. We propose a two-stage Buzz Event Localization and Curation (BELC) workflow: Stage A uses object detection (OD) to detect buzz events and output time-frequency boxes with confidence; a window-geometry-informed consolidation step converts overlapping-window detections into event-level records; and Stage B classifies whether the predicted event represented by each crop is *complete* or *incomplete*, providing a curation step, evaluated on an internal validation set, for filtering low-SNR, truncated, or interrupted detections before duration analysis. Stage A is evaluated against benchmark full-window classifiers, including the Audio Spectrogram Transformer (AST) (Gong et al., 2021) and a YOLOv11 classification baseline adapted to full-window buzz-presence detection (Khanam and Hussain, 2024).

Our main contributions are:

1. A practical two-stage framework for short bioacoustic events in which

Stage A maps raw recordings to event-level time-frequency detections and Stage B classifies consolidated event crops as complete or incomplete for curated duration analysis.

2. A window-geometry-informed event-consolidation and matching protocol that converts overlapping-window OD detections into event-level records while helping preserve overlapping vocalizations as separate event hypotheses.
3. A controlled comparison using a shared task-aligned linear-frequency, high-band spectrogram frontend, showing that Stage A object detection is competitive with full-window AST and YOLO-classification baselines under shared window-level scoring, while uniquely producing explicit event timing and frequency bounds.
4. Illustrative analysis-ready outputs for downstream ecological workflows, including event timing, frequency bounds, duration summaries, and recorder-level data products under explicit post-processing assumptions.

2. Related Work

2.1. Bioacoustic Monitoring and Species Detection

Passive acoustic monitoring (PAM) has become a core approach for biodiversity and behavioral studies because it enables repeated, low-disturbance sampling across large spatial and temporal scales (Sugai et al., 2019). In practice, however, operational PAM systems must handle limited labels, non-stationary background noise, overlap among vocal sources, and domain shift between training and deployment environments (Kahl et al., 2021). These constraints are especially relevant for species- and call-type-specific workflows, where target events are sparse and class boundaries can be subtle.

2.2. Clip-Level Classification for Bird Audio

Recent bird-audio pipelines have shown clip-level use of pretrained and fine-tuned models (including BirdNET and transformer-based approaches) (Kahl et al., 2021; Gong et al., 2021). These systems are typically used for presence/absence questions over fixed windows, but they do not return explicit event boundaries. For ecological tasks that depend on onset, offset, duration, or reliable frequency-band localization (e.g., call-bout structure analysis, or comparisons of frequency-band use with ecological correlates such

as time-of-day, season, or elevation), clip-level predictions alone are often insufficient because multiple events can occur within one window and event timing/frequency placement remain ambiguous. Many general audio classifiers use log-mel frontends designed for broad audio tagging, whereas the present task requires preserving fine frequency structure in a high-frequency band.

2.2.1. BirdNET as a species-level reference model

BirdNET is widely used as a general-purpose multi-label bird-audio classifier in passive acoustic monitoring workflows (Kahl et al., 2021). We considered BirdNET as a contextual reference and ran it on the held-out test audio, but its outputs are not directly comparable to the buzz-specific benchmarks used here.

Two issues limit direct comparison. First, BirdNET produces per-species confidence scores across many trained classes, and species-specific operating thresholds require separate calibration and validation before precision-recall values can be interpreted for a deployment setting (Schwinger et al., 2026). Second, the HLW output node represents species-level acoustic evidence rather than the buzz call type specifically. HLW also produces non-buzz call vocalizations, which are common in the field recordings and were not annotated in our buzz-event datasets; a high BirdNET HLW score in a window without a labeled buzz could therefore reflect a true HLW call rather than a buzz false positive, making false-positive and true-negative counts undefined under the present labels. A separate representational issue is that BirdNET uses a log-mel frontend, which allocates fewer bins per Hz at high frequencies than the linear high-band representation used here and may therefore reduce sensitivity to the frequency structure of HLW buzzes. Thus, BirdNET is best treated here as a species-level contextual model rather than a direct buzz-event benchmark; the quantitative benchmarks in this manuscript focus on models that can be trained and scored directly against buzz-specific window and event annotations.

2.3. Event-Level Detection and Localization

For tasks that require event timing or frequency extent, event-level formulations address the boundary ambiguity of clip-level classification by producing localized predictions over time and, in spectrogram object detection, over both time and frequency. Event-based matching and precision-recall analysis are formalized in the audio event detection literature (Mesaros et al.,

2016) and are increasingly relevant to ecological audio-event detection workflows (Araya-Salas et al., 2023). Strongly annotated bird-audio corpora with species labels and onset/offset times provide individual vocalization intervals and have made time-localized event evaluation operationally feasible in bird-audio workflows (Gómez-Gómez et al., 2023). However, one-dimensional time labels do not provide frequency bounds and may be less suited to cases where concurrent or overlapping vocalizations must be separated into distinct event traces. Two vocalizations can share the same start and end times but occupy different frequency bands, so temporal labels alone cannot distinguish them as separate events. For HLW buzzes, this distinction matters because duration and frequency-band placement are both downstream measurements, and overlapping buzzes from different individuals can occur within the same recording window. The present task therefore uses two-dimensional time-frequency boxes to represent multiple candidate events with confidence and boundaries, supporting transparent post-processing and error analysis under overlap.

2.4. Windowed Object Detection and Box Consolidation

Overlapping-window inference is widely used in computer vision when targets are small relative to the full image or when high-resolution inputs must be processed in manageable tiles. Slicing-aided inference frameworks such as SAHI improve small-object detection by running detectors on overlapping image patches and mapping predictions back to the original coordinate system (Akyon et al., 2022). The resulting duplicated or fragmented boxes are commonly handled with post-processing methods such as non-maximum suppression, Soft-NMS, and weighted box fusion (Bodla et al., 2017; Solovyev et al., 2021). More recent work on window-based object detection explicitly uses detection-window coordinates to suppress or merge redundant partial detections (Kos, 2025). Our consolidation step is related to this windowed-detection post-processing literature, but is adapted to acoustic spectrogram geometry. Although overlap ambiguity is not unique to acoustic detection, it is especially consequential here. Overlapping boxes may reflect duplicate views of the same buzz across adjacent analysis windows, or distinct buzzes that genuinely overlap in time and frequency; overlap alone therefore cannot reliably determine whether detections should be merged. This motivates a consolidation rule that uses window sequence, containment of edge fragments, and the originating analysis window, rather than treating overlap alone as evidence of duplicate detections.

2.5. Measurement Need and Positioning for HLW Buzz Analysis

Existing bird-audio work provides tools for species classification and event detection, and computer-vision work provides methods for post-processing detections from overlapping windows. The measurement capability addressed here is call-type-specific event measurement in noisy field recordings, where buzzes can be incompletely observed, appear in multiple overlapping analysis windows, or overlap with other buzzes and background sounds. HLW buzz analysis requires event-level time-frequency localization, consolidation of duplicate window detections, and completeness-based curation before duration and frequency summaries can be computed. Existing public bird-audio resources and readily available pipelines do not provide this combination of annotations and outputs.

3. Data Collection Overview

3.1. Field Recording Design

AudioMoth acoustic recorders (Hill et al., 2018) were deployed in Sural Bhattori, Himachal Pradesh, India (Fig. 2), within the Himalayan nesting grounds of the Hume’s Leaf Warbler, at 30 sites spanning 3,171 to 3,818 m elevation. Units were spaced at approximately 100 m intervals and mounted 1–3 m above ground. Recordings were collected during May–August in both 2024 and 2025, using four 15-min recording sessions every morning, yielding 2,673.2 hours in total (1,444.7 h in 2024 and 1,228.5 h in 2025). The finalized two-stage pipeline produced 65,280 duration-export records for complete buzzes from this corpus, along with substantial variation in background conditions including wind, non-target bird calls, and site-specific acoustic clutter.

Because this paper focuses on computational methodology, these field recordings define the operational domain for model development and ecological summary analysis.

3.2. Public Audio Sources

Public recordings from Xeno-Canto and iNaturalist (Xeno-Canto Foundation, 2026; iNaturalist, 2026) are used in two roles: (i) early-stage exploration and qualitative understanding of buzz structure, and (ii) out-of-domain stress testing of generalization after model development. We treat these data separately from the held-out field test set used for primary evaluation.

Existing public bird-audio resources provide valuable species-level recordings, but they do not provide the call-type-specific time-frequency event

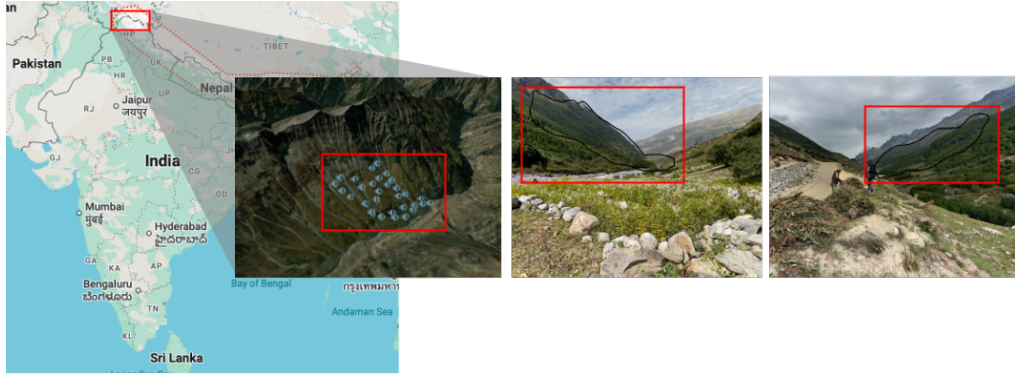


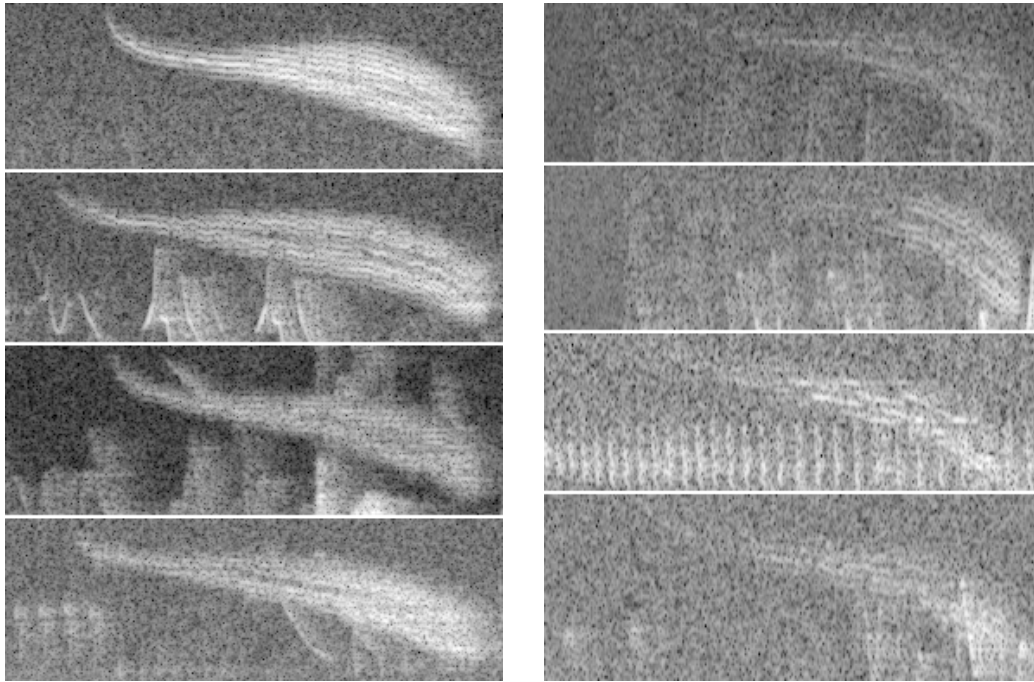
Figure 2: Field recording context for the HLW study area. The composite image combines the study-area location in northern India, an oblique topographic view of the recorder deployment area, and field photographs of the recording sites, with the focal hillside outlined in red.

boxes and complete/incomplete event-quality labels required for duration and frequency-band analysis of HLW buzzes. This study therefore constructs both the recording corpus and the annotation structure needed to evaluate buzz localization and downstream event curation.

3.3. Initial Data Exploration

To guide the design of our detection pipeline, an initial exploratory review was conducted using a combination of public databases and primary field recordings. All recordings were converted to spectrograms using short-time Fourier transform (STFT), and a representative subset was visually examined to assess patterns of vocal activity. This review focused on identifying features that distinguish HLW buzzes from other bird vocalizations and from background sources (e.g., low-frequency environmental sounds, wind, and other non-biological noise), in order to guide detector design. This process confirmed that HLW buzzes possess distinct, repeatable temporal and spectral features suitable for object-detection frameworks.

This analysis showed that the HLW buzz appears as a compact and visually consistent region in the spectrogram (Fig. 1). It is typically represented by a single connected component with a smooth boundary, a shallow concavity, and an internally striped intensity pattern; this interpretation was developed with input from a coauthor with field expertise. These features make it visually distinct from the HLW call, other species’ vocalizations, and background noise. Buzzes were concentrated primarily within the 5–12 kHz



(a) Complete buzz examples.

(b) Incomplete buzz examples.

Figure 3: Representative complete and incomplete buzz examples. Left panel: completely observed buzzes. Right panel: incompletely observed buzzes, where portions of the signal fade into the local noise floor.

frequency band, with durations typically between 1.0–1.5 s. Multiple buzzes could appear in the same frame and overlap in both time and frequency (Fig. 4).

The review also highlighted two observational regimes: fully observed buzzes, where both temporal boundaries and the internal pattern are clearly visible, and incompletely observed buzzes, where weaker portions become indistinguishable from background and only part of the event remains visible. In these incomplete cases, the apparent truncation reflects low local SNR (segments falling below the noise floor) rather than a true biological event boundary (Fig. 3).

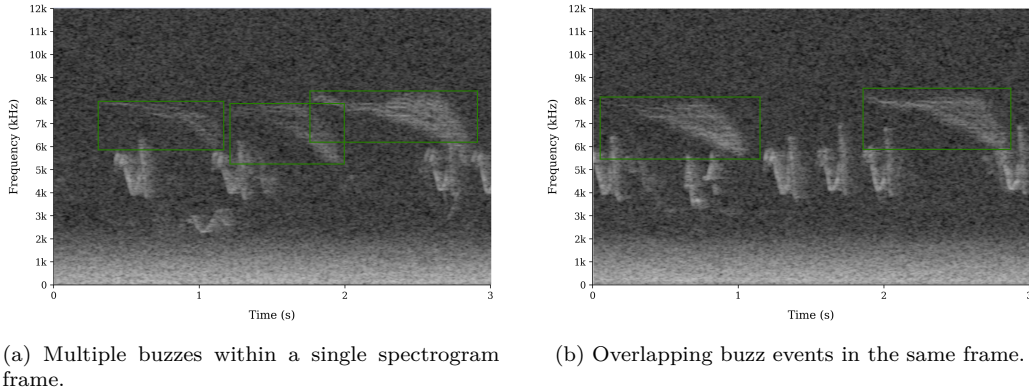


Figure 4: Examples of multiple and overlapping HLW buzzes.

4. Problem Formulation

4.1. Rationale for the Object-Detection Formulation

Ecological analysis requires event-level measurements for each buzz: onset, offset, duration, and occupied frequency band. Recovering these measurements from field recordings is challenging because buzzes occur under variable background noise and can overlap within the same spectrogram window (Section 3.3; Fig. 4). Clip-level presence labels therefore merge distinct events and obscure boundary information. A spectrogram object-detection formulation addresses this directly by predicting multiple time-frequency boxes with confidence, enabling explicit boundaries and one-to-one event matching. Given the compact and repeatable morphology of HLW buzzes, this formulation is well aligned with the signal and analysis goals.

4.2. Addressing Incompletely Observed Buzzes

Incompletely observed buzzes are common: detected buzzes can be only partially visible when weak portions fall below the local noise floor or are transiently masked by other sounds (Section 3.3; Fig. 3). These are still valid event candidates for retrieval, but treating them as complete events biases duration and frequency-bound summaries.

For this reason, completeness filtering is handled as a separate downstream classification problem on normalized event crops. The event localizer is intentionally tuned for high-recall retrieval of candidate buzzes with accurate boundaries, while a dedicated classifier uses its full representational capacity to learn completeness cues (visible onset/offset and uninterrupted

internal pattern). This separation supports independent thresholding and error analysis for retrieval versus curation, improving interpretability and downstream duration reliability.

4.3. Appropriate Time-Frequency Representation

Selecting an appropriate time-frequency representation is a key design decision for this formulation. Mel-scale representations are designed to approximate human auditory perception and therefore allocate progressively fewer frequency bins at higher frequencies, resulting in substantial compression above 1 kHz. This compression reduces frequency resolution and can blur or merge narrowband, high-frequency structures. Because HLW buzzes occur predominantly at high frequencies and exhibit fine-grained spectro-temporal structure, preserving resolution in this range is critical for reliable localization and discrimination. For this reason, the pipeline uses a linear frequency scale rather than a Mel scale.

Figure 5 illustrates this distinction on the same held-out complete buzz. In the BirdNET-style and standard AST frontends, the high-frequency region is increasingly compressed by the mel representation, and the horizontal striation pattern visible in panel (d) is effectively lost in panels (a) and (b); the standard AudioSet AST frontend is additionally band-limited to 8 kHz by its 16 kHz input assumption. The linear-frequency frontend used here preserves uniform spacing in the relevant high-band region and retains the full 4.125–11.625 kHz crop used for Stage A. This band was chosen to comfortably bracket the observed event distribution: across consolidated field detections, the 0.001 quantile of low-frequency bounds was 4.47 kHz and the 0.999 quantile of high-frequency bounds was 10.14 kHz.

The pipeline therefore uses a cropped linear-frequency high-band representation focused on the buzz band. Building on this design rationale, we next formalize the computational components of the pipeline.

4.4. Formulated Task A: Buzz Event Localization

Given audio $x(t)$, Stage A predicts a set of candidate event boxes

$$\mathcal{B} = \{b_i\}_{i=1}^N, \quad b_i = (t_i^{start}, t_i^{end}, f_i^{low}, f_i^{high}, c_i),$$

where t_i^{start} and t_i^{end} are the time bounds of detection i , f_i^{low} and f_i^{high} are its frequency bounds, and c_i is detection confidence.

The role of this component is event retrieval and localization: it identifies where buzz-like events occur in time-frequency space, including cases with overlap or variable background conditions.

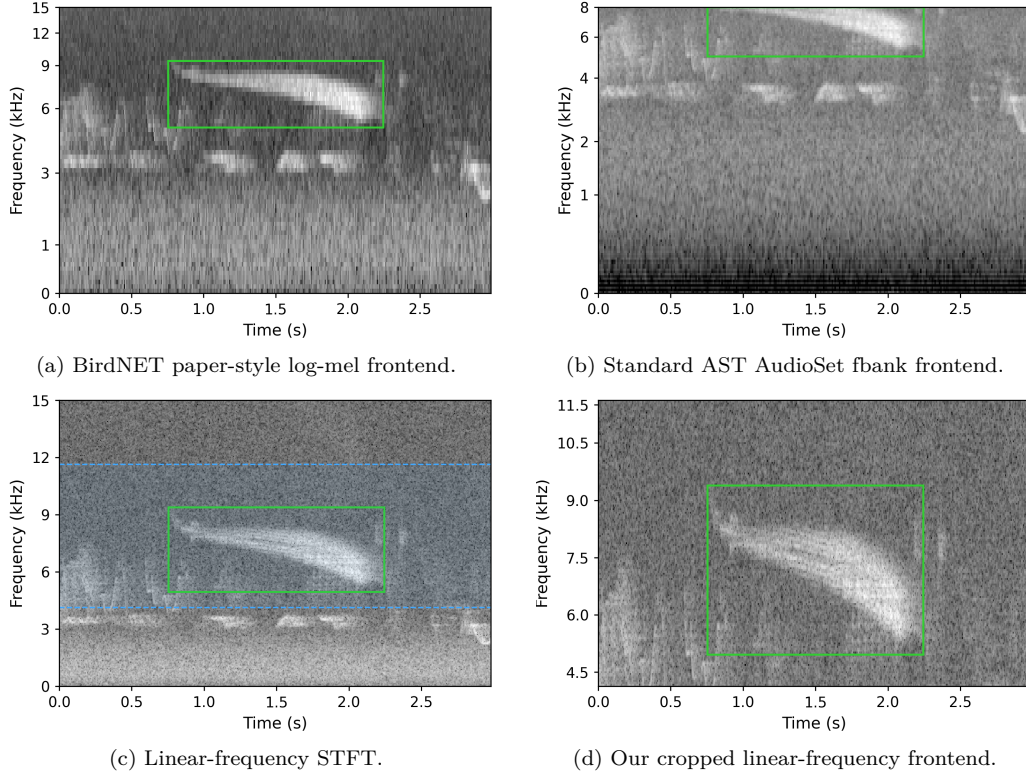


Figure 5: The same strong held-out HLW buzz rendered with BirdNET-style, standard AST, and linear-frequency frontends. Panel (a) shows the BirdNET style frontend, a nonlinear mel-frequency scale spanning 150 Hz–15 kHz. Panel (b) shows the standard AST AudioSet frontend, a nonlinear mel-frequency scale; it assumes 16 kHz audio and therefore spans 20 Hz–8 kHz, so the upper portion of the buzz falls out of range. Panel (c) shows a linear-frequency STFT, with the Stage A analysis frequency band marked in blue. Panel (d) shows our final cropped 4.125–11.625 kHz linear-frequency frontend used in this work. The green box marks the annotated complete buzz in all panels.

4.5. Event Consolidation Across Overlapping Windows

Stage A is applied to highly overlapping spectrogram windows, so a single buzz can produce multiple window-level detections in adjacent windows. The event-consolidation task maps these repeated detections into a set of event-level tracks,

$$\tilde{\mathcal{B}} = \{\tilde{b}_j\}_{j=1}^M,$$

where each $\tilde{b}_j$ represents one candidate buzz event in absolute time-frequency coordinates. The consolidation step must merge repeated views of the same buzz while preserving distinct buzzes that overlap in time and frequency.

4.6. Formulated Task B: Completeness-Based Event Curation

For each event crop z_j derived from consolidated event $\tilde{b}_j$, Stage B predicts

$$\hat{y}_j \in \{\text{complete}, \text{incomplete}\}, \quad p_j = P(\text{complete} \mid z_j).$$

This stage does *not* detect new events. It operates only on consolidated Stage A detections and addresses a different failure mode: bias in duration and frequency summaries caused by incompletely observed events.

Stage A answers *where buzz-like events are*; Stage B answers *whether each predicted event is complete or incomplete for curated duration estimation*. These are distinct tasks and are evaluated separately; downstream duration analysis retains detections passing the selected Stage B completeness threshold.

4.7. Formulated Output: Analysis-Ready Event Records

The consolidated Stage A event records and Stage B curation labels together produce analysis-ready event records. For each retained complete event, the exported record can be written as

$$e_i = (t_i^{\text{start}}, t_i^{\text{end}}, f_i^{\text{low}}, f_i^{\text{high}}, c_i, q_i),$$

where c_i is the Stage A event confidence and q_i is the Stage B completeness confidence. Duration is derived from the consolidated Stage A time bounds:

$$d_i = t_i^{\text{end}} - t_i^{\text{start}}.$$

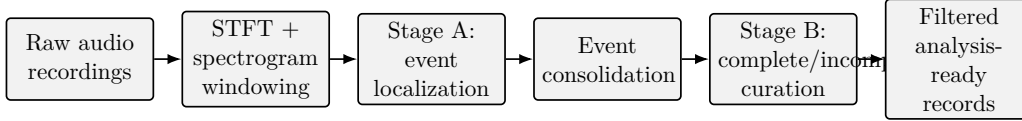


Figure 6: Computational pipeline for buzz event localization and curation.

5. Methods

5.1. Pipeline Overview

The computational workflow used in this study is summarized in Fig. 6 and consists of five steps:

1. Convert audio recordings into overlapping fixed-duration spectrogram windows.
2. Apply Stage A to produce window-level buzz detections with time-frequency boxes and confidence scores.
3. Link detections across adjacent analysis windows to form event-level tracks.
4. Extract standardized event crops and apply Stage B to estimate complete vs incomplete.
5. Apply retention thresholds and export filtered analysis-ready event records.

5.2. Spectrogram Preprocessing and Windowing

The pipeline uses 48 kHz audio and short-time Fourier transform (STFT) front-end processing with a 1024-sample FFT, 256-sample hop length, and a linear-frequency spectrogram representation.

Each analysis window spans 512 STFT frames (2.7467 s), a pragmatic choice that keeps the temporal context close to the 3 s window used in BirdNET while ensuring that the detector input dimensions remain exact multiples of the YOLO stride (32), avoiding padding or resizing in Stage A. The detector input is frequency-cropped to 4.125–11.625 kHz (spectrogram bins 88:248 in this study), flipped along the frequency axis for consistent image orientation, converted to dB relative to per-window maximum amplitude, min-max scaled to 8-bit intensity for normalization, and formatted as native 3-channel 160×512 images for model ingestion. No square resizing or

padding is applied for Stage A; both image dimensions are integer multiples of the detector stride (32).

The analysis window advances by 128 frames (0.6827 s) between consecutive analysis windows, corresponding to 75% overlap. Across the 2024 and 2025 buzz-length analysis outputs, the maximum observed buzz duration was 1.65 s; this window length and stride ensure each buzz is fully contained in at least one analysis window. No front-end denoising is used, as denoising variants were explored but did not improve model performance.

5.3. Stage A Methods: Event Localization and Consolidation

5.3.1. Stage A Model Training

Stage A uses a YOLOv11 object detector (large variant, YOLOv11l) initialized from an MS COCO pre-trained backbone and optimized on the Stage A dataset with controlled geometric/color augmentation. Training was run on three GPUs. The training configuration used for reported experiments was native input size 160×512 , 50 epochs, batch size 48, and initial learning rate 0.001. Applied augmentation magnitudes were a maximum brightness change of 10%, maximum translation of 5%, and maximum scaling change of 5%. The learning rate followed a non-cosine decay schedule from 0.001 to 0.00001 with 3 warmup epochs. Additional optimization settings were momentum 0.937, weight decay 0.0005, warmup momentum 0.8, and warmup bias learning rate 0.1; detector loss gains were 7.5 for box loss, 0.5 for classification loss, and 1.5 for distribution focal loss.

5.3.2. Stage A Inference and Coordinate Mapping

For each analysis-window index n and predicted box $(x_i, y_i, w_i, h_i, c_i)$, Stage A maps normalized coordinates to absolute time and frequency bounds by

$$t_i^{start} = n \cdot \Delta t + T_w \cdot \left(x_i - \frac{w_i}{2}\right), \quad t_i^{end} = n \cdot \Delta t + T_w \cdot \left(x_i + \frac{w_i}{2}\right),$$

and

$$\begin{aligned} f_i^{low} &= f_{max} - \left(y_i + \frac{h_i}{2}\right) \cdot (f_{max} - f_{min}), \\ f_i^{high} &= f_{max} - \left(y_i - \frac{h_i}{2}\right) \cdot (f_{max} - f_{min}). \end{aligned}$$

where Δt is window stride, T_w is window duration, and $(f_{min}, f_{max}) = (4.125, 11.625)$ kHz in this study.

5.3.3. Event Consolidation Across Overlapping Windows

Because analysis windows overlap by 75%, a single buzz often appears in several adjacent windows, including edge-truncated partial views at the start or end of the window sequence (Fig. 7A). We therefore consolidate Stage A detections within each recording into a single event-level prediction for each repeated set of window-level boxes (Fig. 7B).

Consolidation is performed in absolute time-frequency coordinates. We represent each raw detection as

$$b_i = (t_i^-, t_i^+, f_i^-, f_i^+, c_i, w_i),$$

where t_i^-, t_i^+ are absolute start and end times, f_i^-, f_i^+ are lower and upper frequency bounds, $c_i \in [0, 1]$ is detector confidence, and w_i is the source analysis-window index. We treat raw detections as nodes in a within-recording association graph, and each eligible pair of detections from nearby analysis windows defines a candidate link, that is, a possible same-event link considered for consolidation. Candidate links are considered only between detections from different but nearby analysis windows within the same recording; detections from the same analysis window are never linked directly for consolidation:

$$1 \leq |w_i - w_j| \leq G,$$

with fixed G given in Appendix Table A.6.

For a candidate link (b_i, b_j) , let

$$d_i = t_i^+ - t_i^-, \quad \beta_i = f_i^+ - f_i^-,$$

and let $m_t(i)$ and $m_f(i)$ denote the time and frequency centers of b_i . We define duration and bandwidth similarity as

$$\rho_t(i, j) = \frac{\min(d_i, d_j)}{\max(d_i, d_j)}, \quad \rho_f(i, j) = \frac{\min(\beta_i, \beta_j)}{\max(\beta_i, \beta_j)},$$

and normalized center offsets as

$$\delta_t(i, j) = \frac{|m_t(i) - m_t(j)|}{\max(d_i, d_j)}, \quad \delta_f(i, j) = \frac{|m_f(i) - m_f(j)|}{\max(\beta_i, \beta_j)}.$$

To quantify whether a detection is likely truncated by its source window, let ℓ_i and r_i denote the normalized left and right time coordinates of b_i within that window. Its edge proximity is

$$e_i = \text{clip}_{[0,1]} \left(\frac{\eta - \min(\ell_i, 1 - r_i)}{\eta} \right),$$

where η is the fixed edge-proximity scale. For a candidate link, we use the larger edge-proximity value,

$$e_{ij} = \max(e_i, e_j).$$

The pairwise association score is then

$$\begin{aligned} A_{ij} = \text{clip}_{[0,1]} & \left(a_1 \text{IoU}_{2D}(i, j) + a_2 \text{IoU}_t(i, j) + a_3 \text{IoU}_f(i, j) \right. \\ & + a_4 \rho_t(i, j) + a_5 \rho_f(i, j) + a_6 \min(c_i, c_j) + a_7 e_{ij} \\ & \left. - a_8 \min(\delta_t(i, j), 2) - a_9 \min(\delta_f(i, j), 2) - a_{10} \max(|w_i - w_j| - 1, 0) \right), \end{aligned} \quad (1)$$

where the fixed weights $a_1, \dots, a_{10}$ are reported in Appendix Table A.6. Here IoU_{2D} is the full time-frequency box IoU, while IoU_t and IoU_f are one-dimensional IoU on the time and frequency intervals. Thus, A_{ij} is high when two detections overlap well in time-frequency space, have similar extent, lie close in center location, and are plausibly repeated views of the same event across adjacent windows.

A candidate link is retained only if it is *mutual-best* under A_{ij} , meaning that b_j is the highest-affinity partner of b_i and b_i is the highest-affinity partner of b_j . Let

$$m_{ij} = \min \left(A_{ij} - \max_{k \neq j} A_{ik}, A_{ij} - \max_{\ell \neq i} A_{\ell j} \right)$$

denote the mutual-best margin over the second-best alternative on each side. A mutual-best pair is retained only if it passes one of two fixed gates, a *strong-link* gate or a *support-link* gate (Appendix Table A.6). Strong links seed event tracks. Support links are processed afterward and may extend a seeded track, but they are not allowed to merge two already established tracks. To preserve distinct overlapping buzzes as separate event hypotheses, a consolidated track may contain at most one detection from any given analysis window (Fig. 8).

The procedure also treats window-edge fragments explicitly. After the initial linking pass, a near-edge singleton detection p may be absorbed into an existing multi-window track T_k with absolute time-frequency envelope

$$E_k = (t_{E_k}^-, t_{E_k}^+, f_{E_k}^-, f_{E_k}^+).$$

Containment of p within E_k is summarized by area, temporal, and frequency cover quantities C_{area} , C_t , and C_f defined in Eq. A.1. Using those terms, the

singleton-to-track absorption score is

$$S(p, T_k) = g_1 C_{\text{area}} + g_2 C_t + g_3 C_f + g_4 e_p + g_5 c_p + g_6 c_{T_k} - g_7 \min(\delta_t(p, E_k), 2) - g_8 \min(\delta_f(p, E_k), 2), \quad (2)$$

where $c_{T_k} = \max_{i \in T_k} c_i$ is the maximum member confidence of the candidate target track, and $\delta_t(p, E_k)$ and $\delta_f(p, E_k)$ are normalized center offsets computed analogously to the pairwise definitions above after replacing the second box with the track envelope. The fixed weights $g_1, \dots, g_8$ are reported in Appendix Table A.6. A singleton is absorbed only if it is sufficiently edge-adjacent, the target track does not already contain a detection from the singleton’s source window, and the fixed containment, score, and assignment-margin gates are all satisfied (Appendix Table A.6).

Finally, each accepted track T_k is converted to a consolidated event $\tilde{b}_k$. Frequency bounds are estimated by confidence-weighted medians of member-detection boundaries,

$$\tilde{f}_k^- = \text{wmed}\{f_i^- : i \in T_k; c_i\}, \quad \tilde{f}_k^+ = \text{wmed}\{f_i^+ : i \in T_k; c_i\}.$$

Time bounds use the same weighted-median idea, but with edge censoring: starts near the left edge of a source window are excluded from the start vote, and ends near the right edge are excluded from the end vote, so that potentially truncated partial views do not pull the consolidated boundary inward. These censored observations can still preserve observed extent by expanding the final boundary when necessary. The consolidated event confidence is

$$\tilde{c}_k = \max_{i \in T_k} c_i.$$

No event-confidence threshold is applied during consolidation itself; low-confidence singleton events remain available for downstream evaluation and export thresholding. Fixed implementation parameters are summarized in Appendix Table A.6.

5.3.4. Event-Target Matching for Evaluation

For native Stage A event-level localization summaries, consolidated predictions are matched one-to-one to adjudicated reference events within each recording. A consolidated prediction $\tilde{b}_j$ and reference target r_m are considered a candidate same-event pair when they pass a two-axis identity gate: in time, the intervals must overlap or their centers must differ by at most 0.50

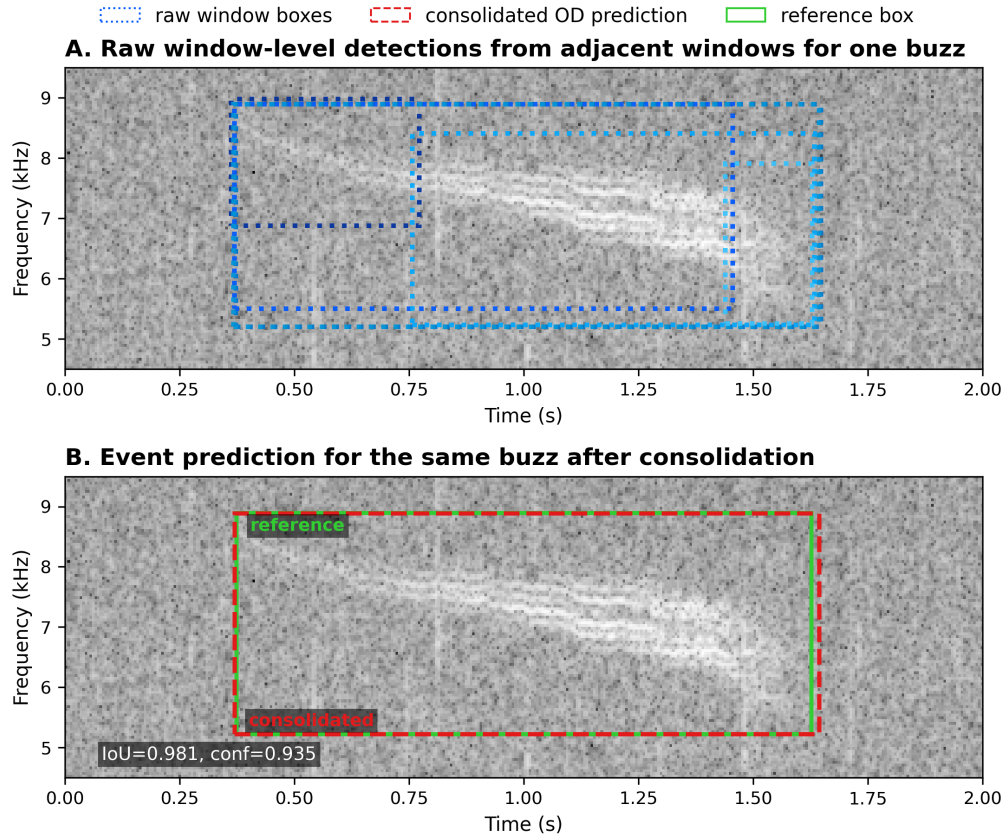


Figure 7: Stage A windowed prediction consolidation for a single buzz. Panels A and B show the same held-out buzz before and after event consolidation: Panel A shows raw window-level boxes produced from adjacent overlapping windows, and Panel B shows the single consolidated event-level prediction with the reference box overlaid.

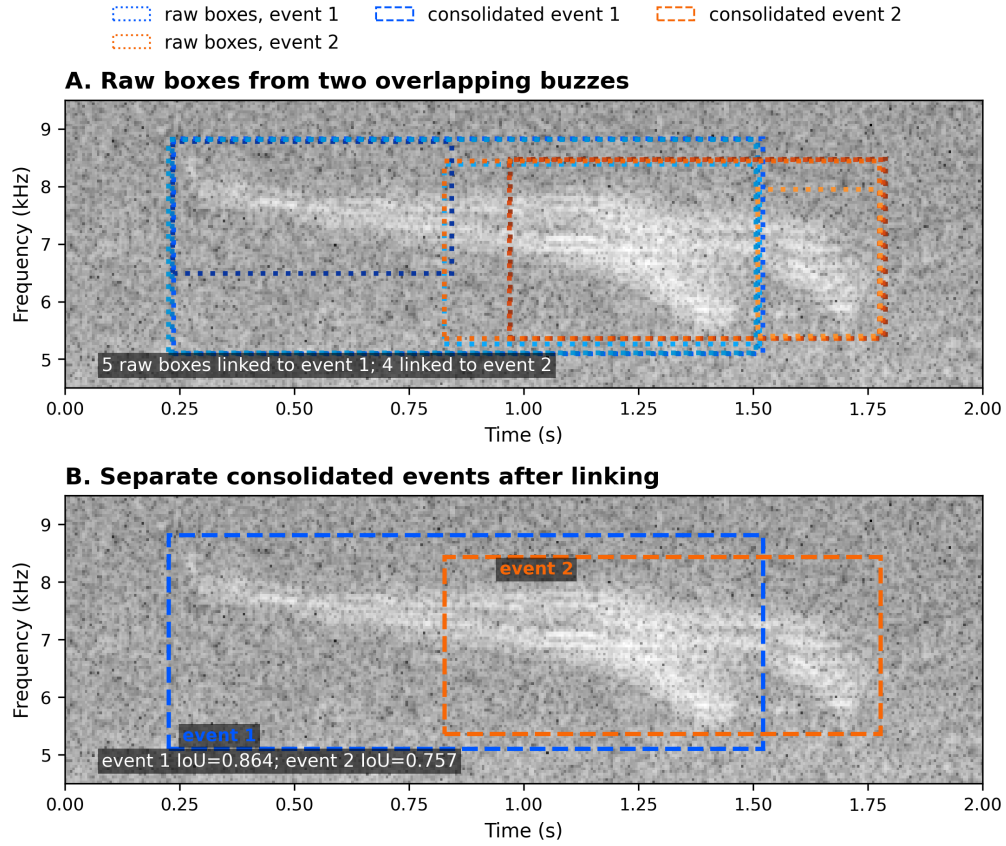


Figure 8: Stage A consolidation for overlapping buzzes. Panel A shows raw window-level boxes grouped by the two event tracks produced by the consolidation procedure. Panel B shows the resulting consolidated event boxes, preserving two separate event hypotheses despite temporal and spectral overlap.

s; in frequency, the intervals must overlap or their centers must differ by at most 1.0 kHz. No hard minimum IoU_{2D} gate is imposed at this stage. Candidate pairs are then assigned with the Hungarian matching algorithm, which selects the best overall one-to-one matching between predictions and reference events. The assignment score is driven mainly by IoU_{2D} , with smaller adjustments for temporal and frequency overlap, center agreement, and event confidence to break near-ties among plausible matches. Assigned targets are counted as true positives, unassigned targets as false negatives, and remaining consolidated predictions as false positives. Under stratum-restricted reporting, predictions that candidate-match only the opposite-stratum ignore set are excluded from false-positive accounting. For precision-recall analyses, predictions are first filtered at each event-confidence threshold and the one-to-one assignment is recomputed at that threshold.

This matching rule separates event retrieval from boundary-localization quality. A prediction may identify the correct buzz even when target-boundary variance or partial-box geometry lowers box overlap. We therefore report IoU_{2D} and marginal time/frequency errors after matching as localization summaries rather than using overlap alone to define a detected event. The identity gate, assignment score, and low-IoU retention rule were developed on Stage A training/validation predictions and then fixed for held-out reporting. Validation sweeps with a conventional IoU_{2D} gate showed that Stage A predictions usually had high overlap with reference labels: with predictions saved at confidence ≥ 0.001 , all 716 validation labels retained a match through an IoU gate of 0.50, with the first loss at 0.55. The final protocol is more tolerant to avoid discarding plausible same-event detections solely because of boundary disagreement. Fixed matching parameters are summarized in Appendix Table A.7.

5.4. Stage B Methods: Completeness Classification of Predicted Buzz Events

Stage B performs event-quality classification on Stage A predicted events. Its role is to estimate whether the predicted buzz event represented by each crop is fully observed; event localization remains defined by Stage A.

5.4.1. Stage B Input Generation from Stage A Predictions

At inference and export time, Stage B operates on consolidated Stage A event tracks rather than raw duplicate window-level boxes. Each consolidated event is converted to a spectrogram crop using the event’s time-frequency bounds and standardized to the fixed crop format used for completeness

labeling (Section 6.4). The same crop-construction procedure is used for Stage B labeling and inference: crops are reconstructed from source audio using the consolidated event bounds, the predicted event is right-aligned in the standardized temporal crop, and the image is resized with preserved aspect ratio and padded to 288×288 .

5.4.2. Stage B Model Training and Validation

Stage B uses a YOLOv11 image classifier (large variant, YOLOv11l-cls) initialized from ImageNet pre-trained classification weights and optimized on the Stage B crop dataset. The training configuration for the run used in this study used an input image resolution of 288×288 , 100 epochs, and batch size 32. Training used an initial learning rate of 0.001 with non-cosine decay to 1% of the initial rate, 3 warmup epochs, momentum 0.937, and weight decay 0.0005. Augmentation was limited to modest brightness, translation, and scale changes. Validation was run on $D_{B\text{-val}}$ each epoch to produce class-level performance summaries. The Stage B checkpoint used for inference was selected as the one with the highest validation peak F1.

5.4.3. Stage B Inference and Retention Rule

At inference time, Stage B outputs a completeness confidence score $q_i = p(\text{complete} \mid z_i)$ for each scored event crop z_i . For downstream field exports, Stage B is applied to consolidated events that pass the validation-derived Stage A event-confidence operating threshold. A detection is retained for downstream duration analysis when its completeness score passes θ_B , where θ_B is the Stage B completeness threshold that attains peak F1 on $D_{B\text{-val}}$ for the selected Stage B checkpoint, then fixed for downstream exports. Detections below threshold are not used in duration summaries but are retained for diagnostic error analysis.

5.5. Duration and Frequency Aggregation for Analysis-Ready Exports

Analysis-ready exports start from event-consolidated Stage A predictions. For field-data exports, consolidated events first pass the validation-derived Stage A event-confidence operating threshold θ_A ; Stage B is then applied to eligible events, and retained duration records are those whose completeness score passes the selected Stage B threshold θ_B :

$$\mathcal{E}_{\text{complete}}(\theta_A, \theta_B) = \{ \tilde{b}_i \in \tilde{\mathcal{B}} : c_i \geq \theta_A, q_i \geq \theta_B \}.$$

For each retained event i , we carry forward time bounds (t_i^{start}, t_i^{end}) , frequency bounds (f_i^{low}, f_i^{high}) , and confidence scores (c_i, q_i) , then compute

$$d_i = t_i^{end} - t_i^{start}, \quad f_i^{center} = \frac{f_i^{low} + f_i^{high}}{2}, \quad \Delta f_i = f_i^{high} - f_i^{low}.$$

These event-level records are linked to recording metadata, including audio file, recorder ID, elevation band, and numeric elevation, to form an analysis-ready table for downstream ecological statistics. Records with missing metadata links, non-positive duration, or invalid frequency bounds are excluded during export.

Primary summaries include duration and frequency distributions by ecological covariates such as elevation, recorder-level descriptive summaries, and counts tracking how consolidated Stage A event records pass through Stage A thresholding, Stage B scoring, and filtering for export as complete events. For recorder-level summaries, retained events are first averaged within recorder, and elevation-band summaries are then computed across recorder-level means rather than individual events.

6. Dataset Development

6.1. Dataset Splits and Evaluation Subsets

All experiments are organized around two data granularities: analysis windows for Stage A and event crops for Stage B. We define five primary train/validation/test splits and one additional disjoint public-audio evaluation subset used for generalization analysis:

- $D_{A\text{-train}}$, $D_{A\text{-val}}$: Stage A (event detection) training and validation sets.
- $D_{B\text{-train}}$, $D_{B\text{-val}}$: Stage B (complete vs incomplete) training and validation sets.
- D_{test} : held-out, manually annotated field recordings for model comparison.
- D_{public} : a disjoint public-audio holdout used for additional generalization analysis, consisting of public-recording submissions unrelated to those used in early training-data development.

The relationships among these partitions and their construction paths, along with the associated dataset counts, are summarized in Fig. 9 and Table 1.

Table 1: Dataset counts for major training and evaluation sets used by the two-stage pipeline.

Stage	Split/Class	Count
Stage A	training windows	34,385
Stage A	validation windows	1,799
Stage A	training buzz boxes	13,718
Stage A	validation buzz boxes	716
Stage B completeness	train complete	17,283
Stage B completeness	train incomplete	12,234
Stage B completeness	val complete	506
Stage B completeness	val incomplete	336
Held-out field test	recordings	54
Held-out field test	final annotated buzz events	1,081
Held-out field test	final complete buzz events	391
Held-out field test	final incomplete buzz events	690
Unseen public audio	files analyzed	38
Unseen public audio	buzz vocalizations	125

6.2. Test-Set Annotation Summary

The primary held-out test set contains 54 field recordings (15 minutes each; 13.5 h total), selected at random from the 2024 field recordings, with 1,081 final HLW buzz-event boxes. Events were labeled in Audacity (Audacity Team, 2026) as time-frequency boxes (start/end time and lower/upper frequency), with non-buzz audio left unlabeled. Each file received two independent researcher annotation passes, followed by an expert adjudication pass to finalize ground-truth labels used only for held-out evaluation. This expert-adjudicated held-out set supports Stage A event-detection and localization evaluation.

Complete/incomplete tags used for Stage A stratum reporting follow the same expert-guided visibility criteria used for Stage B crop labeling, defined below. However, Stage B completeness model performance is evaluated only on its internal validation split rather than on an independently expert-adjudicated test set. Timeline-level inter-annotator agreement, quantified with Krippendorff’s α over time bins including silence intervals, was 0.92. Detailed adjudication mechanics, including weighted-vote logic and agreement implementation details, are documented in Appendix B.

Review of preliminary Stage A outputs prompted a final annotation-QC

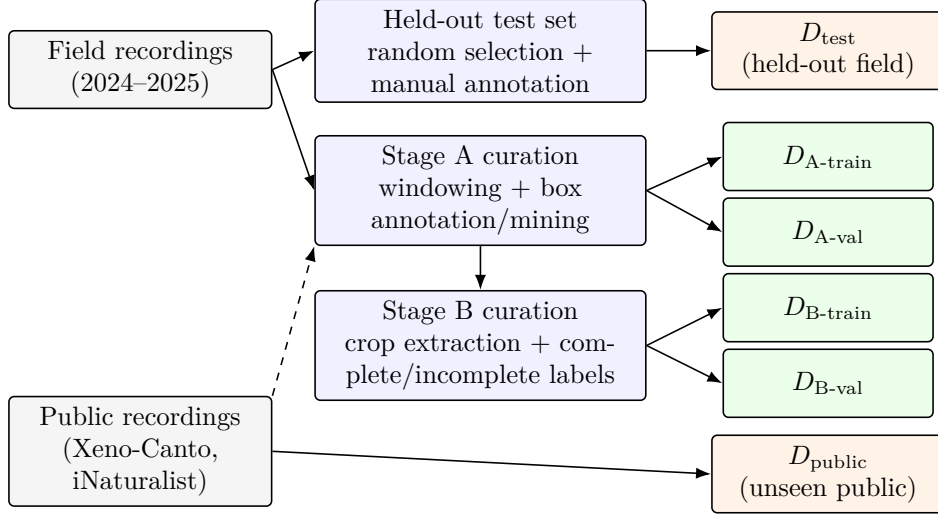


Figure 9: Data flow and partitioning used in this study. Solid arrows indicate primary construction paths. Dashed arrow indicates limited use of other public submissions in early seed curation. D_{public} denotes a disjoint public-audio holdout and was not itself used for training, model selection, or seed curation.

pass focused on duplicate or misplaced boxes, overlapping-event separation, and boundary placement. This step was important for fair comparison across all Stage A models because the same reference events determine window-level labels, stratum membership, precision-recall accounting, and native Stage A event-target matching. The final held-out event labels contain 391 complete buzzes and 690 incomplete buzzes; analyses run before this pass were used only for error analysis.

6.3. Training Data Curation: Buzz Event Localization

Training data were built iteratively. A small initial dataset combining examples from Xeno-Canto, iNaturalist, and field recordings was manually annotated in CVAT.ai (CVAT.ai, 2026) using time-frequency bounding boxes around buzz events, and used to train an initial detector. That model then mined candidate positives/negatives from larger unlabeled field audio. Candidate windows were sampled across detector-confidence strata and elevation bands from non-test recordings, then manually verified and box-corrected in CVAT. Recurrent false-positive patterns were explicitly retained as hard negatives. Repeating this loop expanded representation of background conditions, overlapping events, non-stationary noise, and acoustically similar hard

negatives.

For Stage A baseline comparisons, a clip-level binary dataset was also derived from Stage A box annotations using the window-label construction protocol defined in Section 7.2; the resulting benchmark comparison is described in Section 7.5.

6.4. Training Data Curation: Buzz Crop Completeness Labeling

Stage B uses a binary crop-level label scheme: *complete* versus *incomplete*. To build this corpus, candidate buzz crops are generated from non-test field detections. Here, each crop is the time-frequency rectangular region of the spectrogram corresponding to a predicted buzz box. Each detection is converted to a standardized crop with a fixed temporal width of 288 frames, with the predicted event right-aligned in the crop and left-side context or padding added as needed. Temporal padding is sampled from local non-buzz regions to preserve realistic background texture. The time dimension is preserved at 288 frames, and the frequency dimension is padded with gray background values to form 288×288 classifier inputs.

Stage B crops were uploaded to CVAT.ai for manual binary tagging. The tagging rules were developed with guidance from a coauthor with field expertise, and tagged crops were then organized into train/validation splits (counts in Table 1). A crop is labeled *complete* only when both conditions hold:

- temporal containment: both onset and offset are visible within the crop;
- continuous structure: the buzz appears as one continuous structure rather than an edge-truncated fragment.

Crops that fail either condition are labeled *incomplete*. Representative labeled crop examples are shown in Fig. 3.

7. Experiments

7.1. Evaluation Sets and Data Leakage Controls

We evaluate models on a held-out, manually annotated field test set (D_{test}) and a separate unseen public-audio set (D_{public}). Training and mining are performed on non-test recordings. Test recordings are excluded by filename during mining and inference to prevent data leakage. Thresholds used for downstream exports and retained-event decisions are selected on development/validation data. Cross-model Stage A benchmark comparisons

use average precision (AP) as the primary summary metric; peak F1 from test-set threshold sweeps is retained only as a secondary diagnostic and is not interpreted as a deployment operating threshold.

From D_{test} , we construct three reporting strata for Stage A: (i) all buzzes + background, (ii) complete buzzes + background, and (iii) incomplete buzzes + background. Primary cross-model benchmark reporting focuses on the all-buzz and complete-buzz strata, because downstream duration/frequency analysis retains only completely observed events. In stratum-restricted analyses, the opposite buzz stratum is treated as ignore (excluded from TP/FP/FN accounting), while background negatives are retained. This stratification serves two complementary goals: all-buzz performance quantifies general buzz retrieval under field conditions, while performance on complete buzzes isolates the retrieval behavior most relevant to downstream duration/frequency analysis. The stratum of incomplete buzzes remains useful as a secondary diagnostic for distant, occluded, or otherwise low-SNR events, but is not emphasized in the primary benchmark table.

7.2. Experimental Setup and Matching Procedure

For cross-model Stage A comparison, all Stage A models are evaluated under a common window-level protocol so that retrieval can be compared on the same analysis windows and with the same precision-recall procedure. Let $W = \{w_k\}$ denote this full test-window universe (2.7467 s windows, 0.6827 s stride). Under this protocol, each model must provide a single confidence score $s_k \in [0, 1]$ for each window w_k . This is the native output format of the classifier baselines, but object detection (OD) instead produces zero, one, or multiple time-frequency boxes within a window. We therefore convert OD output to a single window score by taking the maximum box confidence among detections in that window. This conversion is used only for the shared cross-model benchmark and discards localization information, which is evaluated separately in the native event-level analyses.

Let R_g denote the reference events included for reporting stratum g (all buzzes, complete buzzes, or incomplete buzzes). We report two complementary window-labeling criteria. The *complete-within-window* criterion asks whether a complete reference event is contained in the analysis window:

$$y_k = 1 \iff \exists r_j \in R_g \text{ such that } t(w_k)^{\text{start}} \leq t(r_j)^{\text{start}} \wedge t(w_k)^{\text{end}} \geq t(r_j)^{\text{end}}.$$

The *minimum-overlap* criterion asks whether a window contains a meaningful

partial view of a reference event:

$$y_k = 1 \iff \exists r_j \in R_g \text{ such that } O_t(w_k, r_j) \geq d_{\text{match}},$$

where $O_t(w_k, r_j)$ is temporal overlap duration between window w_k and reference event r_j , and $d_{\text{match}} = 0.2$ s. This threshold was selected during analysis review because buzz structure begins to become visually identifiable at approximately this amount of temporal overlap. Windows with no overlap with any reference event in the reporting universe are negatives. Windows that overlap a reference event but do not satisfy the selected positive criterion are treated as *ignore* and excluded from TP/FP/FN accounting. The complete-within-window criterion emphasizes complete-event availability within a frame, whereas the minimum-overlap criterion tests model behavior on boundary windows where only part of the event is visible.

We implement scoring with the following steps:

1. **Construct common evaluation windows.** Enumerate all quarter-step windows across test recordings and assign each model a score per window (missing model outputs are set to 0).
2. **Assign positives, negatives, and ignore windows.** Apply the selected window-labeling criterion for positives. Mark non-qualifying overlap windows as ignore; keep non-overlap windows as negatives.
3. **Apply stratum masks.** For complete-only and incomplete-only strata, treat the opposite buzz stratum as ignore and exclude from scoring.
4. **Threshold sweep and confusion accounting.** For each threshold θ , set $\hat{y} = 1[s \geq \theta]$ on the non-ignored windows and compute TP/FP/FN, then record $(P, R, F1)$.

At each threshold θ , precision, recall, and F1 are:

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}, \quad F1 = \frac{2PR}{P + R}.$$

7.3. Stage A Metrics (Detection)

Stage A cross-model comparison uses the shared window-level protocol defined in Section 7.2. We sweep confidence to generate precision-recall curves and use AP as the primary model-comparison metric. For the main

benchmark table, uncertainty is summarized with paired 95% recording-level bootstrap confidence intervals for the AP difference between OD and each classifier baseline, preserving dependence among overlapping windows from the same recording. Peak F1 from the same threshold sweeps is treated only as a secondary descriptive summary. For stratum-restricted analyses, the opposite buzz stratum is treated as ignore and excluded from TP/FP/FN accounting; these conditional scores quantify retrieval within the selected stratum rather than complete-vs-incomplete discrimination. Primary main-text reporting focuses on all buzzes + background and complete buzzes + background.

Because Stage A is an object detector, we also report native OD event-retrieval precision-recall summaries after event consolidation against the final held-out event boxes described in Section 6.2. These OD-specific event-level retrieval metrics are reported separately from the shared classifier comparison.

7.4. Stage B Metrics (Completeness Classification)

Stage B is evaluated as a binary classifier on complete vs incomplete labels using the validation split $D_{\text{B-val}}$. This split is used for internal model selection and threshold calibration rather than as an independently expert-adjudicated test set. We report validation precision, recall, F1, and area under the PR curve (AUCPR). The Stage B model checkpoint is selected by highest validation peak F1. The retained-event threshold θ_B is then set to the confidence threshold that attains peak F1 for that selected checkpoint, and fixed for downstream duration analysis.

7.5. Stage A Benchmark Comparison: OD vs AST and YOLO Classification

A central experimental objective is to test whether framing buzz detection as object detection can match strong window-level classifiers on detection quality while providing native event-localization outputs. We therefore compare Stage A OD directly against two baselines trained on the same Stage A windows:

- **AST benchmark:** the Audio Spectrogram Transformer, used here as a benchmark audio detection model on full analysis windows.
- **YOLO-classification baseline:** a computer-vision classification baseline adapted to full-window buzz-presence detection.

For comparability with Stage A OD inputs, AST was trained on the same 2.7467 s analysis windows and the same 4.125–11.625 kHz linear-frequency spectrogram representation used for Stage A, rather than the standard AudioSet-style AST frontend. This departs from native AST preprocessing, which assumes 16 kHz audio and a log-mel representation; in our setting, that frontend truncates most buzzes above 8 kHz and compresses the remaining high-frequency structure. Preliminary experiments with the native AST frontend performed poorly, so we used the task-aligned frontend for the benchmark comparison. Both classifiers output a single confidence score $p(\text{buzz present} \mid \text{window})$ per window. All three models are trained and evaluated under the same split constraints and shared window-level protocol in Section 7.2, ensuring a fair formulation-level comparison.

7.6. Native Event-Level Localization Capabilities and Metrics

In addition to OD event-retrieval summaries, we report native Stage A time/frequency localization diagnostics on matched events, including localization summaries and qualitative localization examples and error analyses (Section 7.7.3). Event-level localization evaluation follows the matching procedure in Section 5.3.4, with one-to-one assignment rerun at each event-confidence threshold because the eligible consolidated predictions change with threshold. These diagnostics evaluate the boundary geometry needed for downstream duration and frequency summaries rather than window-level buzz presence alone.

7.7. Results

7.7.1. Stage A Detection Results

Native Stage A event-level evaluation measures how well consolidated predictions recover individual annotated buzz events and their time-frequency boundaries on the final held-out event annotations.

Stage A performed strongest on complete buzzes, the stratum most directly relevant to downstream duration exports (Table 2). The row for complete buzzes achieved $\text{AP} = 0.9995$ and peak $\text{F1} = 0.9911$; at the peak-F1 threshold for complete buzzes, the model recovered 388 of 391 complete buzzes with four false positives. Qualitative review indicated that the remaining false positives were a mix of borderline buzz-like cases and hard negatives with close buzz-like features. Representative error cases at this operating point for complete buzzes are shown in Fig. 11. Performance remained strong for all buzzes combined ($\text{AP} = 0.9600$, peak $\text{F1} = 0.9299$), while performance

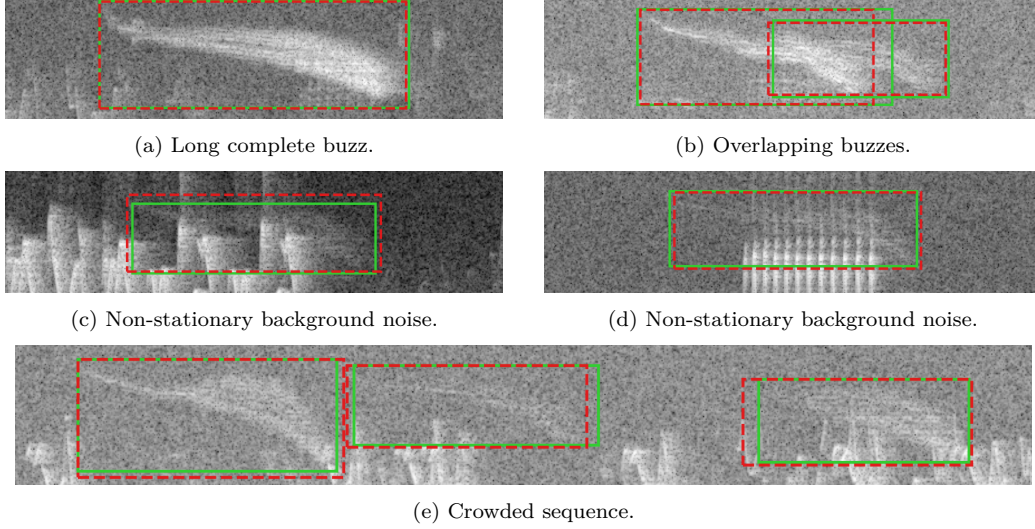


Figure 10: Held-out Stage A event-localization examples after overlapping-window consolidation and one-to-one event matching. Solid green boxes mark reference events and dashed red boxes mark matched consolidated predictions. In all panels, time increases from left to right and frequency from bottom to top. Panel (a) shows a long complete buzz (event confidence 0.92, $\text{IoU}_{2D} = 0.98$). Panel (b) shows overlapping buzzes resolved as separate events (0.93/0.91; 0.90/0.90). Panels (c) and (d) show predictions under non-stationary background noise (0.82/0.80; 0.83/0.92). Panel (e) shows a crowded sequence of nearby buzzes (0.93/0.82/0.92; 0.92/0.91/0.89).

on incomplete buzzes was lower, consistent with the greater ambiguity of distant, truncated, masked, or otherwise low-SNR events.

Representative held-out event predictions are shown in Fig. 10, including localization of a long complete buzz, overlapping buzzes, non-stationary background noise, and crowded sequences.

7.7.2. Stage A Comparison with Window-Level Classifiers

Table 3 provides a controlled comparison asking whether the richer object-detection formulation sacrifices basic buzz-retrieval performance when all Stage A models are evaluated under the same coarse window-level protocol in Section 7.2. AST and YOLO-classification use their direct window confidence scores, whereas OD uses the maximum detection-box confidence in each window. We report AP under the two window-labeling criteria defined there: fully within window containment and partial within-window overlap ($d_{\text{match}} = 0.2$ s).

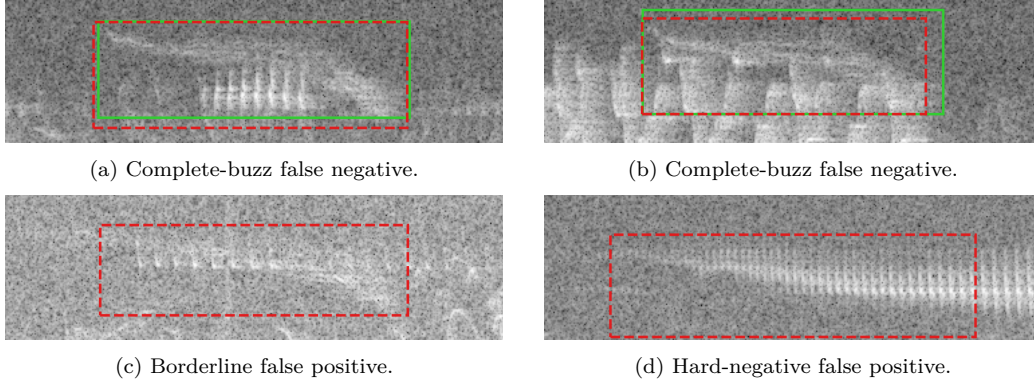


Figure 11: Representative Stage A OD error cases at the held-out peak-F1 reporting threshold for complete buzzes ($\theta = 0.838761$). Panels (a) and (b) show false negatives on complete buzzes in which the model still produced well-localized consolidated event predictions (dashed red) below this reporting threshold; the consolidated event confidences were 0.791 and 0.798, respectively, and reference boxes are shown in solid green. Panel (c) shows a borderline buzz-like case in which non-stationary interference overlaps a faint buzz-like trace, yielding a false positive consolidated event. Panel (d) shows a hard negative case with no annotated buzz event, but with high-band structure resembling a buzz and therefore producing a false positive prediction. In all panels, time increases from left to right and frequency from bottom to top.

Under the fully within-window criterion, OD is modestly lower than both classifier baselines for all buzzes, while complete-buzz AP is near ceiling for all three models. Under the partial within-window criterion, OD and AST are similar for all buzzes, whereas YOLO-classification is somewhat higher overall. For complete buzzes, the setting most relevant to partially visible but fully observed events, OD exceeds AST and is effectively tied with YOLO-classification. Taken together, these comparisons show that Stage A OD remains competitive under a shared window-level benchmark while uniquely returning native event timing and frequency bounds.

7.7.3. Native Stage A Time-Frequency Localization Outputs

Figure 12 and Table 4 summarize localization quality for matched true-positive consolidated events using the event-confidence threshold selected from the all-buzz peak-F1 sweep, with localization errors reported separately for all buzzes, complete buzzes, and incomplete buzzes. These summaries evaluate boundary quality after event consolidation and one-to-one event-target assignment, rather than raw window-level boxes. Because this fixed threshold differs from the peak-F1 threshold for complete buzzes in Table 2,

Table 2: Native Stage A event-level performance on the final held-out test set after sequence-aware event consolidation and one-to-one event-identity matching. Peak F1 and the threshold column are reported from a full event-confidence threshold sweep as post hoc diagnostic summaries, not as deployment-threshold estimates. For stratum-restricted rows, the opposite buzz stratum is treated as an ignore set. The shaded row indicates the evaluation stratum most directly relevant to downstream duration exports; the field-export confidence gate is selected separately from validation data.

Stratum	N_+	Ignored	AP	Peak F1	Precision	Recall	Threshold	TP/FP/FN	Med. IoU _{2D}
All buzzes	1,081	0	0.9600	0.9299	0.9330	0.9269	0.272790	1002/72/79	0.8987
Complete buzzes	391	690	0.9995	0.9911	0.9898	0.9923	0.838761	388/4/3	0.9337
Incomplete buzzes	690	391	0.9297	0.8986	0.9046	0.8928	0.272790	616/65/74	0.8623

Table 3: Primary Stage A model comparison on the manually annotated test set using average precision (AP). Background negatives are retained in all reported strata. Bracketed intervals denote paired 95% recording-level bootstrap confidence intervals for OD relative to each classifier baseline, using resampling over the 54 held-out recordings. FW = fully within window; PW = partially within window (minimum-overlap criterion with $d_{\text{match}} = 0.2$ s).

Context	OD AP	Δ vs AST	Δ vs YOLO
PW All	0.9234	0.0049 [−0.012, 0.016]	−0.0227 [−0.049, −0.011]
PW Complete	0.9874	0.0178 [0.008, 0.031]	0.0001 [−0.006, 0.003]
FW All	0.9594	−0.0219 [−0.046, −0.010]	−0.0262 [−0.055, −0.011]
FW Complete	0.9984	−0.0015 [−0.009, 0.000]	−0.0015 [−0.009, 0.000]

matched counts are not expected to be identical across the two summaries. Matched positives show high overlap overall, with stronger localization for complete buzzes than for incomplete buzzes; for complete buzzes, the median IoU_{2D} was 0.934 and the median absolute duration error was 0.0202 s.

7.7.4. Stage B Completeness Classification Results

This subsection reports Stage B complete/incomplete classification quality on its held-out validation split, separate from detection performance. This validation set was developed with expert guidance, and the metrics below evaluate the selected classifier within that validation framework rather than on a separate external adjudication set. Metrics are reported for the selected checkpoint at the validation peak-F1 threshold.

Representative labeled crops are shown in Fig. 3, illustrating the visual structure of both “complete” and “incomplete” examples used in Stage B evaluation.

Table 4: Stage A OD localization error for matched true-positive events at the all-buzz peak-F1 event-confidence threshold. Temporal errors are median absolute errors in seconds; frequency errors are median absolute errors in Hz; IoU median and IQR are reported separately.

Metric	All buzzes	Complete buzzes	Incomplete buzzes
Matched TP	1002	391	611
IoU _{2D} median	0.899	0.934	0.863
IoU _{2D} IQR	0.824–0.941	0.900–0.958	0.776–0.918
Onset (s)	0.0178	0.0091	0.0334
Offset (s)	0.0181	0.0139	0.0209
Duration (s)	0.0328	0.0202	0.0513
Low freq. (Hz)	80.6	62.8	93.8
High freq. (Hz)	67.5	59.6	79.7
Center freq. (Hz)	57.9	44.5	66.0

Using the selected checkpoint and the peak-F1 validation threshold $\theta_B = 0.530306$ on $p(\text{complete})$, Stage B achieved precision 0.938856, recall 0.940711, F1 0.939783, and AUCPR 0.983888 on $D_{B-\text{val}}$ (842 validation samples: 506 complete, 336 incomplete). At θ_B , the confusion counts were 476 / 31 / 30 / 305 (TP / FP / FN / TN), corresponding to accuracy 0.927553.

7.7.5. Generalization on Unseen Public Audio

An important question is whether the finalized pipeline transfers beyond the field-recording conditions and curated development data used in the main study. We therefore evaluate it on unseen public recordings as an out-of-domain generalization check, using different Xeno-Canto submissions from those involved in early training-data development and spanning different recordists, devices, and locations. Figure 13 provides curated qualitative examples, and the quantitative summary below focuses on whether the pipeline can recover completely observed buzzes under these heterogeneous public-audio conditions. We evaluated Stage A on 38 Xeno-Canto clips containing 125 annotated buzz targets.

We focus here on the Stage B completely observed subset defined on the public targets. In these unseen public recordings, the pipeline recovered all 69 completely observed buzzes (69/69; recall = 1.000). Review at the summary threshold $\theta = 0.805$ found no spurious off-target buzz localizations; the remaining four excess event counts reflected duplicate representations of already-detected buzzes.

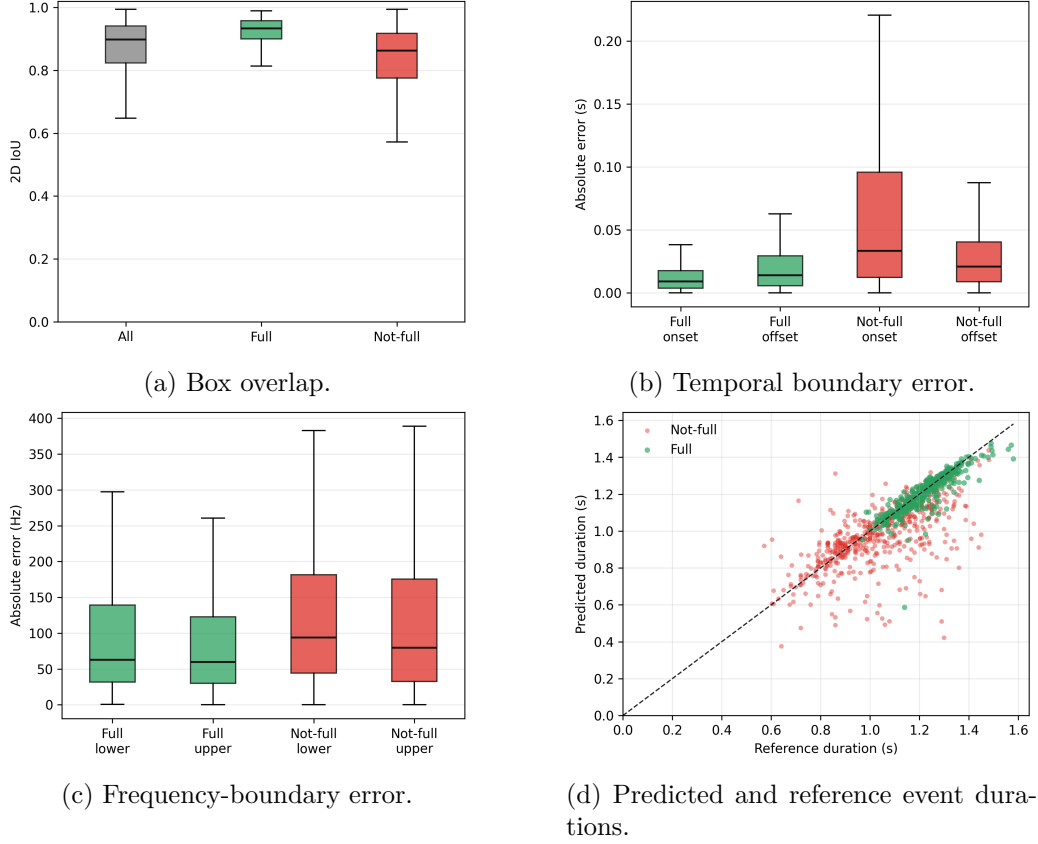


Figure 12: Stage A OD localization distributions for matched true-positive consolidated events at the all-buzz peak-F1 event-confidence threshold.

7.8. Analysis-Ready Outputs for Downstream Ecological Analysis

This section illustrates analysis-ready summaries produced by the finalized two-stage pipeline. It does not present new ecological inference claims.

7.8.1. Completeness Curation for Exported Durations

Stage B completeness curation defines which consolidated Stage A event tracks are eligible for duration analysis. Across 2024 and 2025, the pipeline produced 200,737 consolidated Stage A event tracks (134,701 in 2024; 66,036 in 2025), of which 65,280 were retained as duration-export records for complete buzzes after applying the validation-derived Stage A operating threshold ($\theta_A = 0.539$), Stage B completeness curation, and export data-cleanliness checks for metadata linkage and valid bounds (39,105 in 2024; 26,175 in

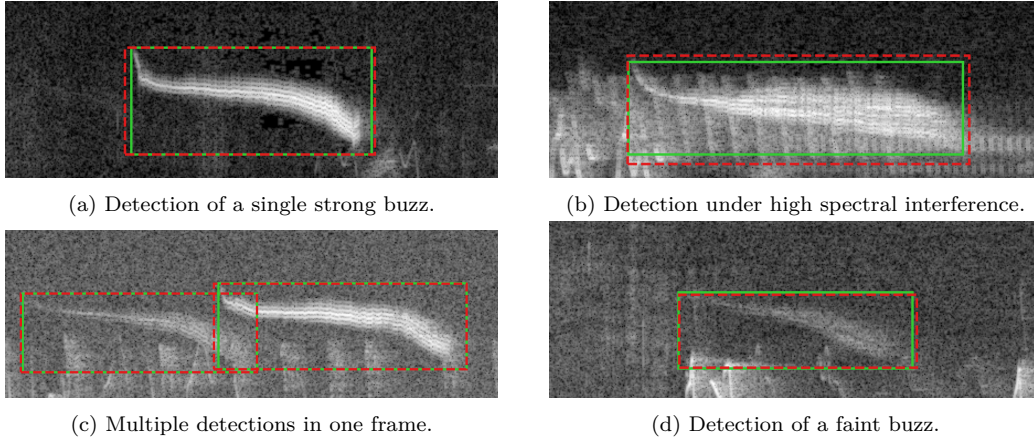


Figure 13: Representative Stage A detections on unseen Xeno-Canto recordings. Solid green boxes show annotated references and dashed red boxes show consolidated event predictions; horizontal axis is time (s).

2025). This validation-derived export gate is separate from the post hoc test-set threshold sweeps reported in Table 2. Event tracks not retained for export either do not pass the Stage A/Stage B gates or fail export-cleanliness checks; they are excluded from duration summaries so the retained data product contains confident, completely observed events with auditable metadata.

7.8.2. Example Data Product: Duration and Frequency by Elevation

All summaries in this section are derived from the duration-export retained event set unless otherwise stated. We present these outputs as an example data product for downstream ecological modeling, not as a formal ecological test. Figure 14 summarizes the export pathway and the kinds of temporal and spectral summaries produced by the pipeline, and Table 5 gives the corresponding recorder-level 2025 elevation-band summary. Export counts are shown for both field seasons, while detailed duration and frequency summaries are shown for 2025 as a single-year worked example to keep the illustration readable.

For the elevation summaries, recorder is the primary descriptive unit: retained events are averaged within recorder, and recorder-level means are then summarized by elevation band. This reflects how buzz duration versus elevation would be analyzed downstream, because events from the same recorder are not independent observations of elevation. Event-level summaries still describe the exported event population, but recorder-level summaries reduce

pseudo-replication and prevent high-activity recorders from dominating apparent elevation patterns.

Table 5: Illustrative 2025 recorder-level summary for duration-export records for complete buzzes. Event counts report retained events, while duration and center-frequency entries are mean $\pm$ SD across recorder-level means within each elevation band.

Elevation band	Recorders	Events	Median events/rec.	Duration (s)	Center freq. (Hz)
Low	7	11,921	451.0	1.2823 ± 0.0810	7115.0 ± 166.1
Medium	8	11,875	1545.5	1.3082 ± 0.0461	7118.1 ± 100.3
High	7	2,379	59.0	1.1335 ± 0.0921	7057.5 ± 152.7

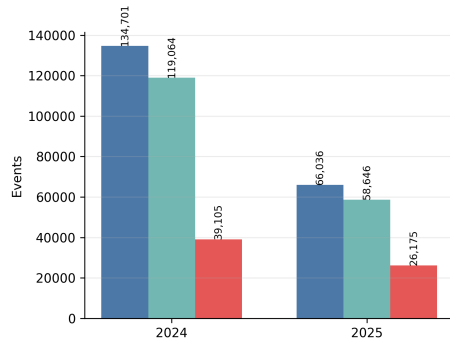
These summaries should be interpreted with the pipeline settings in view: the retained event set depends on the Stage A event-confidence operating threshold, event-consolidation parameters, the Stage B completeness threshold, and export data-cleanliness checks for metadata linkage and valid bounds. Downstream ecological analyses should treat these settings as documented filtering choices and audit them as needed in sensitivity checks.

8. Discussion

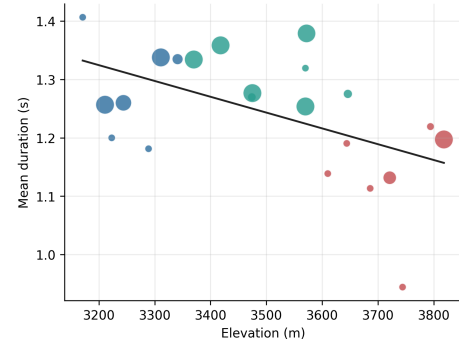
The central methodological result is that separating event localization, event consolidation, and event-quality curation produces a clearer path from noisy field recordings to ecological measurements. This decomposition makes retrieval, boundary placement, duplicate-window handling, and completeness curation separately auditable rather than collapsing them into a single performance score.

The AST and YOLO-classification baselines provide a useful window-level benchmark: OD is modestly lower under complete-within-window scoring, but under the partial-overlap criterion it exceeds AST and is effectively tied with YOLO-classification for complete buzzes. The more important distinction for this measurement task is output structure. A full-window classifier could serve as an initial screen, but a workflow requiring duration or frequency bounds would still need a second localization step to convert positive windows into event-level time-frequency records, introducing additional annotation, calibration, and error-propagation issues. The object-detection formulation keeps that localization native to Stage A while remaining competitive for detection.

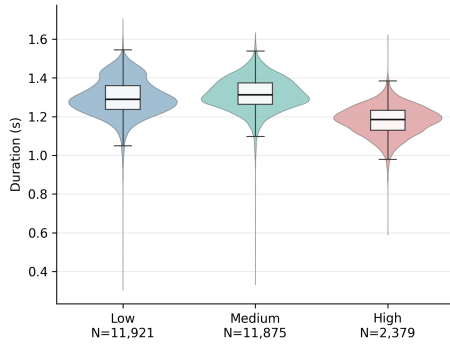
Event consolidation is the bridge between tiled spectrogram inference and ecological records. Standard box-merging logic is not sufficient in this setting



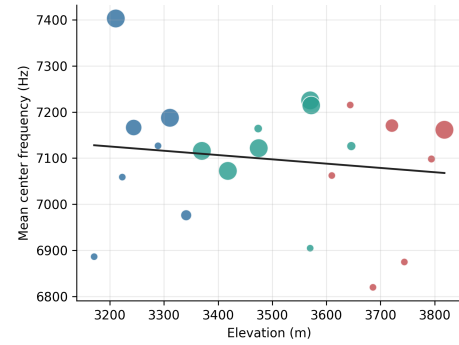
(a) Pipeline export counts.



(b) Recorder-level duration means versus elevation, 2025.



(c) Event duration distributions by elevation band, 2025.



(d) Recorder-level center-frequency means versus elevation, 2025.

● Low-elevation recorders ● Medium-elevation recorders ● High-elevation recorders

Figure 14: Pipeline-derived analysis-ready data product. In Panel A, blue, teal, and red bars denote consolidated Stage A event tracks, Stage-B-scored tracks, and duration-export complete buzzes, respectively. In Panels B and D, point area is proportional to the number of retained events for each recorder. Panel C shows the retained 2025 export set as descriptive event-level context rather than as the inferential sampling unit. Panels B–D are illustrative summaries of exported data products rather than formal ecological inference tests.

because overlap has two meanings: repeated detections of the same buzz in adjacent windows, and genuinely distinct buzzes that overlap in time and frequency. The consolidation procedure is therefore a domain-specific adaptation of windowed object-detection post-processing. The one-to-one event-target matching protocol serves a similarly specific role: it defines event identity for retrieval evaluation, while localization quality is reported afterward through IoU and boundary-error summaries. Together, these choices prevent plausible same-event detections from being discarded solely because of boundary disagreement, while helping preserve overlapping vocalizations as separate event hypotheses.

The ecological summaries are intentionally presented as data products rather than formal ecological inference. They demonstrate that the pipeline can export recorder-linked event records with duration and frequency fields. For questions such as buzz duration versus elevation, recorder-level summaries may be more appropriate than raw event-level summaries because events from the same recorder share sampling context and are not independent observations of elevation.

8.1. Limitations and Reproducibility

This study relies on a sequence of fixed processing choices and thresholded post-processing steps. While this enabled rapid methodological iteration, it increases reproducibility burden and motivates future consolidation into a standardized execution protocol. Additional limitations include domain specificity (single species/call type), annotation effort requirements, dependence on post-processing thresholds and consolidation parameters, observed error modes such as faint overlaps, structured noise that mimics buzz texture, and annotation uncertainty near event boundaries, and the absence of an independently expert-adjudicated Stage B completeness test set. Stage A baseline comparisons are harmonized with a shared window-level scoring protocol, but native Stage A event-localization outputs still involve formulation-specific post-processing for downstream ecological products. We therefore document fixed parameters, identify sensitivity checks needed for downstream ecological inference, and retain appendix documentation of adjudication and parameter settings.

9. Conclusion

We presented a two-stage computer-vision pipeline on spectrograms for HLW buzz analysis, where Stage A localizes events, overlapping-window detections are consolidated into event-level records, and Stage B provides a completeness filter selected and evaluated on the Stage B validation set to support duration estimation. A central result is that Stage A object detection remains competitive with AST and YOLO-classification full-window baselines, exceeds AST and is effectively tied with YOLO-classification in AP for complete buzzes under the partial-overlap criterion most aligned with downstream use, and additionally returns event-level time and frequency bounds required for downstream ecological analysis. This decomposition provides a cleaner mapping from computational outputs to ecological statistics by separating detection, consolidation, curation, and downstream measurement.

More broadly, the study supports a practical template for PAM applications that require event-level time-frequency localization: evaluate localization and quality filtering as separate tasks, then export structured outputs for downstream ecological analysis with explicit assumptions and sensitivity checks.

Data Availability

The model training and test datasets used in this study will be made available upon publication.

Acknowledgements

To be completed.

Appendix A. Parameter Tables

Appendix Tables A.6–A.8 summarize the fixed Stage A consolidation, event-target matching, and Stage B configuration settings used in this study.

For edge-singleton absorption, containment is evaluated for a singleton box p relative to a candidate track envelope E . Area cover is $|p \cap E|/|p|$, time cover is the fraction of the singleton’s temporal extent contained in

Table A.6: Stage A event-consolidation parameter summary (fixed configuration used in this study).

Parameter	Value
Raw prediction ingestion	Window-level Stage A boxes with confidence $c \geq 0.001$
Candidate window gap	Up to 3 analysis-window steps
Association features	2D, time, and frequency IoU; center distances; duration/bandwidth similarity; minimum confidence; time-edge proximity; window gap
Association score weights	0.45 2D IoU; 0.14 time IoU; 0.14 frequency IoU; 0.10 duration ratio; 0.08 bandwidth ratio; 0.05 minimum confidence; 0.06 edge support; subtracting 0.05 center-time distance, 0.04 center-frequency distance, and 0.03 per extra window gap
Strong-link gate	affinity ≥ 0.72 , 2D IoU ≥ 0.55 , minimum confidence ≥ 0.25 , mutual-best margin ≥ 0.04
Support-link gate	affinity ≥ 0.62 , 2D IoU ≥ 0.35 , minimum confidence ≥ 0.10 , mutual-best margin ≥ 0.03 , window gap ≤ 2
Support-link requirements	Candidate edge: maximum confidence ≥ 0.70 or edge support ≥ 0.50 ; track merge: one side has maximum confidence ≥ 0.70 or an existing strong edge
Edge-proximity scale	0.08 of analysis-window duration
Edge-singleton absorption	enabled for near-edge singleton detections mostly contained in established tracks
Absorption edge requirement	Edge proximity ≥ 0.80 for singleton detections
Absorption containment thresholds	area cover ≥ 0.70 , time cover ≥ 0.55 , frequency cover ≥ 0.55
Absorption score threshold	score ≥ 0.72 with margin ≥ 0.08
Absorption score weights	0.55 area cover; 0.15 time cover; 0.15 frequency cover; 0.08 edge value; 0.04 singleton confidence; 0.03 target-track confidence; subtracting 0.04 center-time distance and 0.03 center-frequency distance
Overlap-preservation constraint	At most one detection from a given analysis window per event track
Event-boundary fusion	Time: edge-censored confidence-weighted median with extent preservation; frequency: confidence-weighted median
Event confidence	Maximum member-detection confidence

E , and frequency cover is the fraction of the singleton’s frequency extent contained in E :

$$\begin{aligned} C_{\text{area}} &= \frac{|p \cap E|}{|p|}, \\ C_t &= \frac{|[t_p^{\text{start}}, t_p^{\text{end}}] \cap [t_E^{\text{start}}, t_E^{\text{end}}]|}{|[t_p^{\text{start}}, t_p^{\text{end}}]|}, \\ C_f &= \frac{|[f_p^{\text{low}}, f_p^{\text{high}}] \cap [f_E^{\text{low}}, f_E^{\text{high}}]|}{|[f_p^{\text{low}}, f_p^{\text{high}}]|}. \end{aligned} \tag{A.1}$$

Appendix B. Annotation and Agreement Details

This appendix records annotation quality-control procedures for transparency and reproducibility. These procedures are supporting workflow details and are not presented as a novel contribution of this paper.

Workflow summary.

- Two independent researcher passes were completed on the held-out test recordings.
- An expert adjudication pass reviewed candidate buzz events and boundary placement for the held-out event-detection test set.
- Agreement was quantified over timeline bins including silence intervals.

Adjudication rule (supporting detail). The annotation protocol used weighted consolidation: expert annotations carry higher decision weight than individual non-expert annotations. Events supported by the expert and at least one researcher are accepted; events unsupported by the expert are not promoted solely by non-expert agreement. We include this here as provenance of label curation logic rather than as a methodological claim.

Agreement statistic. Krippendorff’s alpha for the finalized test annotation timeline was 0.92, computed over timeline bins including silence intervals.

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Table A.7: Stage A event-target matching protocol for native event-level evaluation on the final held-out target labels.

Parameter	Value
Matching scope	Within recording
Development data	Stage A training/validation predictions and labels; fixed before final held-out reporting
Prediction unit	Consolidated Stage A event track
Reference unit	Final adjudicated time-frequency target box
Eligible predictions	All consolidated predictions are available to the matching sweep before applying the current event-confidence threshold
Candidate gate	Time overlap or center-time distance ≤ 0.50 s; frequency overlap or center-frequency distance ≤ 1.0 kHz
Assignment method	One-to-one linear-sum assignment (Hungarian matching)
Assignment score	$\text{IoU}_{2D} + 0.25\text{IoU}_t + 0.25\text{IoU}_f + 0.25C_{\text{small}} - 0.10\delta_t - 0.10\delta_f + 0.001c$, where C_{small} is smaller-box coverage, δ_t and δ_f are normalized center-distance penalties, and c is event confidence
IoU interpretation	Localization metric reported after matching; not a hard event-identity gate
True positives	Assigned target-prediction pairs
False negatives	Unassigned reference targets
False positives	Eligible unassigned consolidated predictions; for stratum-restricted rows, candidate matches to the opposite-stratum ignore set are excluded
PR-threshold handling	Filter predictions at each confidence threshold, then rerun assignment
Calibration diagnostic	Train/validation IoU sweep; validation retained 716/716 labels at a conventional 0.50 gate with predictions saved at confidence ≥ 0.001
Low-IoU handling	Plausible same-event matches below $\text{IoU}_{2D} = 0.50$ are retained and evaluated through localization-error summaries

Table A.8: Stage B parameter summary (configuration used in this study).

Parameter	Value
Classifier input size	288×288
Crop construction	Reconstructed from source audio using consolidated event bounds; predicted event right-aligned in the standardized temporal crop
Temporal target crop length	288 frames
Image standardization	Aspect-preserving resize followed by padding to 288×288
Padding strategy	Local temporal background during labeling; mean/gray padding for square image standardization
Classifier model family	YOLOv11 (classification)
Checkpoint selection	Highest validation peak F1 on $D_{B\text{-val}}$
Completeness score	$p(\text{complete})$
Completeness threshold	$\theta_B = 0.530306$ from validation peak F1
Export eligibility	Applied after the validation-derived Stage A event-confidence operating threshold
Training epochs	100
Training batch size	32
Output classes	complete, incomplete

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